Annotating the food domain across different contexts

In this work, we propose annotation guidelines to capture food and its uses with a historical perspective. These guidelines are inspired by Frame Semantics, therefore, we briefly introduce here the concept that we borrowed from this theory for the present annotation guidelines.

Lexical Units and Frame Elements

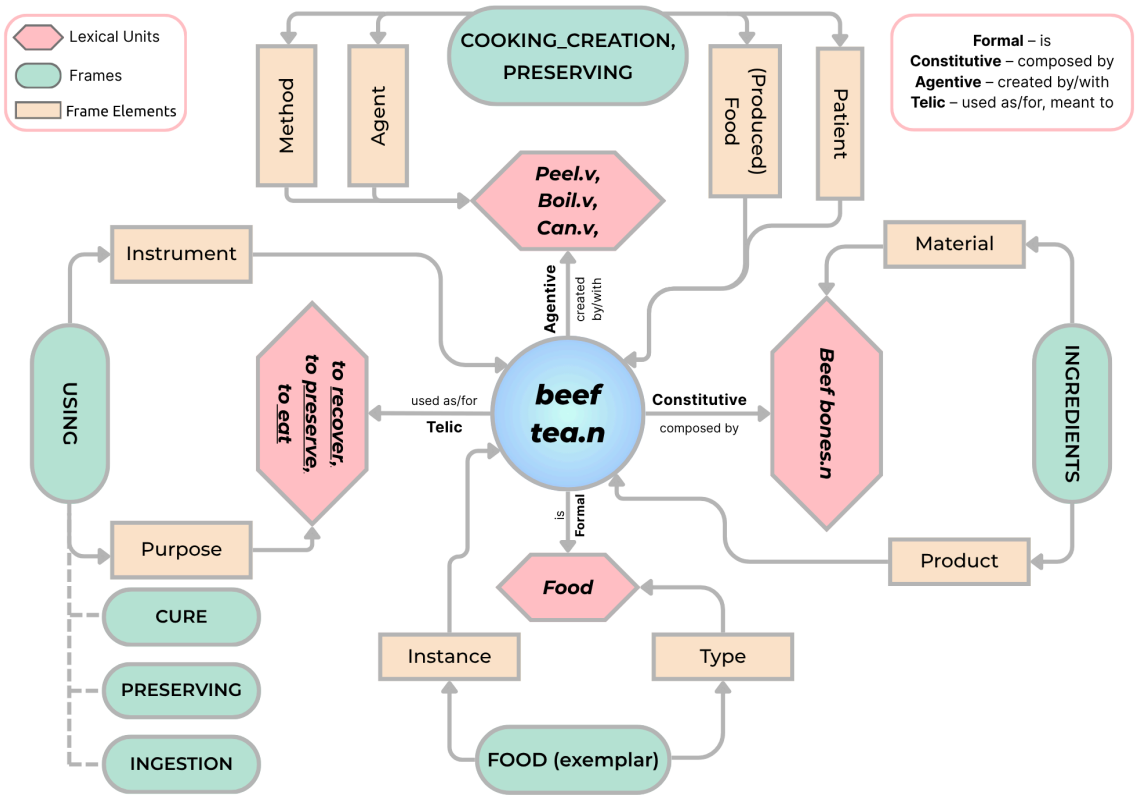
Frame Semantics is a linguistic theory for which the meaning of events, entities, and situations can be represented through **Frames**, schematic representation of scenarios that involve participants, objects, and conceptual roles, known as **Frame Elements**. A **Frame** can be seen as a shared mental scenario or experience that people automatically activate to understand a word or expression. It provides the background knowledge needed to interpret who is involved, what happens, and how the situation normally unfolds. Each frame is evoked by a set of **Lexical Units (LU)**, which may be a single word or a multiword expression, defined as the pairing of a lemma with a specific meaning.

For instance, interpreting the verb *buy* in the sentence *'Daniel bought a new car'* requires a cognitive model, a mental representation of commercial exchange. In this sentence, *bought* is the LU that evokes the **Commerce_buy** frame, whose corresponding FEs are **Buyer** (*Daniel*) and **Goods** (*a new car*). By contrast, in a sentence such as *'Daniel really buys anything you tell him'*, the verb *buy* does not refer to the concrete act of purchasing an object, but to the metaphorical meaning of *believing*. Here, despite its identical linguistic form, *'buy'* can be distinguished from the previous example as it evokes a different frame, **Fall_for**, which involves a distinct set of FEs compared to the commercial scenario. This example is to show that each meaning of a word can activate a different frame, and that polysemous words (words with same form and different meanings) can be distinguished according to the frame they trigger. In essence, Frame Semantics views word meanings as understood in relation with the broader situations or experiences to which these words belong, and each frame includes characteristic roles, such as who acts, what is acted upon, and with what outcome.

Food beyond consumption

We are interested in capturing food from an historical perspective as well as considering any conceptualization of food which goes beyond its consumption as a source of nutrition. The purpose is to represent food in its realizations with a focus on its uses, manipulation and purposes different from nutrition. We express this through an

adaptation of Frame Semantics and relations among frames as illustrated in Fig. 2. In the following paragraph, we present the characterization of each frame through the specific semantic relations and their annotation process. These relations are used only to illustrate the different perspectives and uses through which food is conceptualized. The annotation, however, focuses exclusively on frames, and not on the relations between them.



Annotation

The annotation of food and its uses is done with INCEpTION. The first step of the annotation is the identification of the Lexical Unit of the frame **Food**, namely every word referring to a food entity (e.g., apple, soup, meat, etc.). The standard definition of food in the Oxford English Dictionary is "any nutritious substance that people or animals eat or drink in order to maintain life and growth" and serves here as a baseline for identifying food. But, we do not restrict our notion of food to its act of consumption. For instance, substances that are normally ingested as nourishment but can also be applied externally, such as on the skin to treat a wound, are included in our conceptualisation of food.

In other words, we consider food as any substance typically used for nutrition, even when it is employed for non-ingestive purposes. The annotator has to select the word and choose the correct label *Food_LU* which stands for Lexical Unit of Food. The annotator has to annotate just the word referring to the food without the article, unless the article or any other element is indispensable to define the entity.

Once you identified the LU, the possible frames which are "activated" are:

A. **FOOD**: entity food → relation IS

Food frame is activated by a LU of **Food** through the ***IS*** relation (formal). This Frame is defined in FrameNet as words referring to items of food, and it is considered the exemplar frame in this schema. This means we see the LUs of the frame as the starting point to activate all the possible relations with the other frames. For instance the word 'beef tea' in the flowchart is a LU of the **Food** frame. The **Food** frame consists of *Food (LU)*, and the frame elements *Constituent_parts*, *Descriptor,* and *Type* (which is included in *Descriptor* in our guidelines). In addition there are negative, positive, and neutral annotations for Descriptors and 2 FE from **Measures** frame. The final frame in our schema is then structured as:

Food [FOOD_LU]: corresponds to the LUs

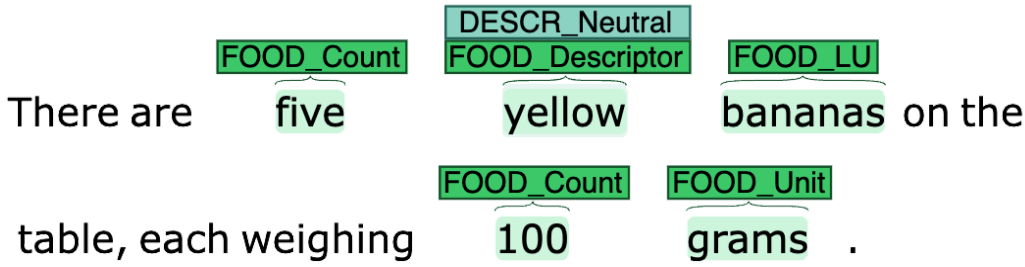
(1) *Constituent_parts* [FOOD_Constituent_Part]: a part of the *Food_LU*;

(2) *Descriptor* [FOOD_Descriptor]: a characteristic or description, as well as a particular *Type* of the *Food_LU*; which can be (1) *Positive* [DESCR_Positive]; (2) *Negative* [DESCR_Negative]; (3) *Neutral* [DESCR_Neutral].

(3) *Count* [FOOD_Count]: the number of units of the measured *Food_LU*;

(4) *Unit* [FOOD_Unit]: the standardized quantity of *Food_LU* in terms of which other

quantities of the same kind are measured.



B. **INGREDIENTS**: entity food → relation COMPOSED BY

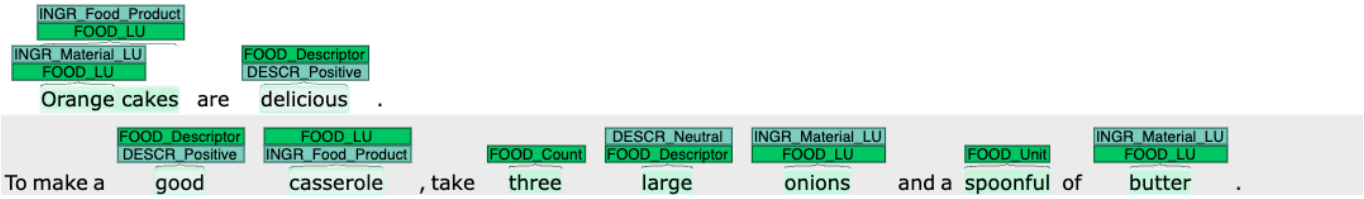
Ingredients Frame is activated by any LU expressing any food material which is used to create a food product through the ***COMPOSED BY*** relation. This Frame is a *Material* used to make a *Product*, in this case any food which is used to create another. In our annotation guidelines, we kept the LUs and Frame Elements *Material*, and *Product*.

<u>Core</u>:

(1) *Material* [INGR_Material_LU]: The entity which is used to make a Product.

(2) *Product* [INGR_Food_Product]: The entity or substance that is created out of the Material, in this case the food.

Notice that the annotator has to double annotate the FOOD_LU both as the frame LU and the specific FE of the other activated frame (in this case *Product_Food* of **Ingredients**). In the list of labels in INCEpTION, the FE which corresponds to the exemplar frame LU is marked with *_Food* label, for instance the FE of **Ingredients**, *Product* is *Product_Food* in our annotation schema.



C. **COOKING_CREATION/APPLY_HEAT**: entity food → relation CREATED BY/WITH

Unlike the previous two, these frames model events rather than entities. In this case, we consider three frames in the manipulative(created with) and agentive (created by)

characterisations, namely COOKING_CREATION, APPLY_HEAT, and PRESERVING. These frames describe food and meal preparation and manipulation, including any type of manipulation performed, explicitly or implicitly, by an agent on the food, either to create or to modify it. For the COOKING_CREATION frame, we consider only the two core FEs, namely *Cook* and *Produced_Food*. While APPLY_HEAT focuses primarily on the process of handling and transforming the ingredients by heating it, COOKING_CREATION emphasises the edible entity resulting from the process. We aggregate these two perspectives under COOKING_CREATION, which covers a broader range of food manipulation verbs. Here, food annotated as *Produced_Food* can represent either the raw food to which the process is applied (as in APPLY_HEAT) or the final product resulting from the manipulation (as in COOKING_CREATION).

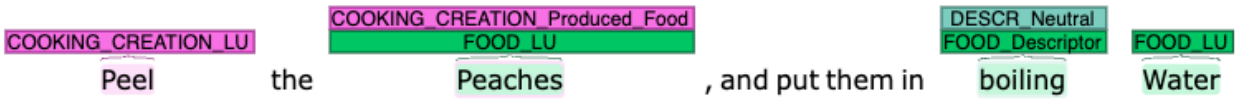
<u>COOKING_CREATION/APPLY_HEAT</u>:

Cook [COOKING_CREATION_Cook]: The person who prepares the Produced_Food.

Produced_Food [COOKING_CREATION_Produced_Food]: The Produced_Food is the result of a Cook's efforts.

The Cooking Creation LUs (with some of them borrowed by **Applying Heat**) are all those verbs which evoke the cooking process. A non-exhaustive list is provided below:

bake.v, baking.n, barbecue.v, blanch.v, boil.v, boiling.n, braise.v, broil.v, broiling.n, brown.v, char.v, coddle.v, concoct.v, cook up.v, cook.n, cook.v, cooking.n, deep fry.v, fix.v, fry.v, frying.n, grill.v, grilling.n, make.v, melt.v, melting.n, microwave.v, parboil.v, plank.v, poach.v, preparation.n, prepare.v, put together.v, roast.v, roasting.n, roast.v, roasting.n, saute.v, scald.v, scorch.v, sear.v, simmer.v, simmering.n, singe.v, steam.v, steaming.n, steep.v, stew.v, stewing.n, toast.v, toasting.n, whip up.v



PRESERVING: entity food → relation CREATED WITH

For the PRESERVING frame, we focus on the role of the preserved *Patient*. In our annotation, the *Patient* must be realised as food to trigger this frame, thus becoming *Food_Patient,* which is preserved to prevent deterioration, potentially using a *Medium* that can itself be food. While food can be annotated as either *Patient* or *Medium*, our primary focus in this realisation is on the Food-to-Patient relation, which captures the manipulation aspect. Nonetheless, as we are also interested in cases where food functions as a preserving tool, its purpose-related realisation (in technical terms, the *telic* role[1]), presented in the next section, includes instances where it acts as a preserving medium.

In this frame an *Agent* preserves a *Patient* (food) in order to prevent it from decaying. A

[1] **Telic/telicity** refers to an action that expresses a **purpose** or is oriented toward a specific goal.

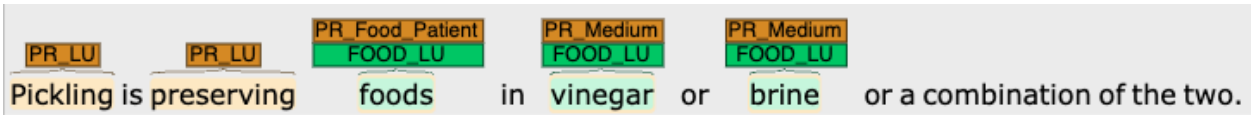
Medium may be used.

(1) *Agent* [PR_Agent]: The Agent is the person performing the intentional act that leads to the preservation.

(2) *Medium* [PR_Medium]: Medium is the substance in which the Patient might be submerged or treated to be preserved.

(3) *Patient* [PR_Food_Patient]: The Patient is organic or non-organic matter that undergoes preservation.

can.v, cure.v, dry.v, pickle.v, preservation.n, preserve.v, salt.v, tinned.a, smoke.v



D. **USING**: prototypical frame for purpose → relation USED AS/USED FOR/MEANT TO

This last frame differs from the others in that it provides a prototypical structure for a set of frames that share the same purpose-oriented configuration, each capturing a different use or purpose of food. This relation refers to the specific function or intended purpose of an object, namely what it ***IS MEANT TO*** or what ***IS USED AS/FOR***. In the case of food, this dimension of telicity is encoded in the core FEs of the frame USING, which is taken here as a reference model for examining how use-driven meanings of food are structured. These elements, recurring across the selected frames, support the conceptualisation of the various uses of foods.

We used the core FEs of USING as a reference point to identify the shared FEs in the selected frames. Specifically, the relevant FEs are: *Agent*, the individual or entity using the instrument to achieve a goal; *Instrument*, the entity used by the Agent to fulfill their purpose which in our scheme is the FOOD_LU; and *Purpose*, which identifies the specific goal or reason for using the Instrument (here food), which in our scheme is represented by the specific LUs of the selected telic frames (e.g., CURE_LU, PR_LU, etc.).

For **Using**, the core FEs of this Frame are:

(1) *Agent*: The Agent uses the Instrument to achieve their Purpose.

(2) *Instrument*: An entity directed by the Agent that serves to bring about the Purpose of the Agent.

(3) *Purpose*: This FE identifies the purpose for which an Agent uses the Instrument.

This frame has a list of lexical units which we provide below:

application.n, apply.v, employ.v, employment.n, exercise.v, operate.v, operation.n, operational.a, use.n, use.v, utilisation.n, utilization.n, utilise.v, utilize.v

The frames activated by the telic relation are:

E. **CURE** entity food → relation USED FOR/AS, MEANT TO

The purpose of food is to treat specific afflictions. It concerns a *Healer* treating or curing an *Affliction* (such as an injury, disease, condition, or pain) affecting a *Patient*, specifically when a food item is mentioned as a particular *Treatment* or *Medication*. We focus solely on the core FEs, and we aggregate Medication and Treatment into the single FE *Food_Treatment*. This choice reflects the fact that food is conceptually closer to *Treatment*, while not excluding its potential overlap with *Medication*.

The FEs are:

(1) *Affliction* [CURE_Affliction]: The Frame Element Affliction is generally the NP Object of a verb, frequently incorporating the Patient as a possessor.

(2) *Body_part* [CURE_Body_Part]: The Body_part is the specific area of the Patient's body which is treated.

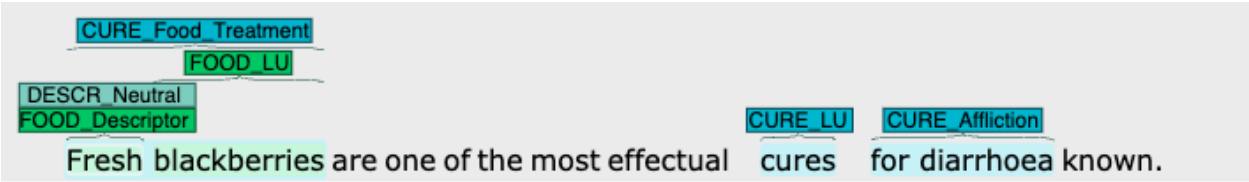
(3) *Healer* [CURE_Healer]: The Healer represents anyone who treats or cures the Patient.

(4) *Treatment/Medication* [CURE_Food_Treatment]: The ingested, applied, injected, etc. substance designed to cure the Patient.

(5) *Patient* [CURE_Patient]: This is the sufferer of the injury, disease or pain and can occur as an NP Object in this frame. However, Patient may not be expressed as a separate Frame Element but frequently occurs as the possessor of the Affliction. The Patient may therefore not be tagged in a sentence although explicitly identified.

A list of LUs is provided below:

alleviate.v, alleviation.n, curable.a, curative.a, curative.n, cure.n, cure.v, ease.v, heal.v, healer.n, incurable.a, improve.v, nurse.v, palliate.v, palliation.n, palliative.a, palliative.n, rehabilitate.v, rehabilitation.n, rehabilitative.a, remedy.n, remedy.v, resuscitate.v, soothe.v, therapeutic.a, therapist.n, therapy.n, treat.v, treatment.n, subdue.v



F. **INGESTION** relation MEANT TO

In this frame, the purpose of food is its most common one: consumption. The frame formalises this by representing an Ingestor who consumes a food or drink item, the *Ingestibles* (here, *Food_Ingestibles*), an action that involves placing the item in the mouth

with the intent of delivering it to the digestive system.

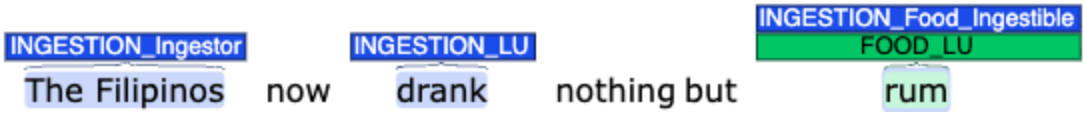
The core FEs of this frame that we considered in this study are:

(1) *Ingestibles* [INGESTION_Food_Ingestible]: The Ingestibles are the entities that are being consumed by the Ingestor.

(2) *Ingestor* [INGESTION_Ingestor]: The Ingestor is the person eating or drinking.

The LUs are all those verbs or noun which refer to the act of eating and drinking

breakfast.v, consume.v, devour.v, dine.v, down.v, drink.v, eat.v, feast.v, feed.v, gobble.v, gulp.n, gulp.v, guzzle.v, have.v, imbibe.v, ingest.v, ingestion.n, lap.v, lunch.v, munch.v, nibble.v, nosh.v, nurse.v, put away.v, put back.v, quaff.v, sip.n, sip.v, slurp.n, slurp.v, snack.v, sup.v, swig.n, swig.v, swill.v, tuck.v



G. **PRESERVING** relation USED AS/FOR, MEANT TO

We have already seen the PRESERVING frame for food as *Patient* of the preserving methodology. In its telic realization, the purpose of food is that of being the substance used to preserve another item from decay, so the food is a *Medium* to preserve something else. The LUs are the same already listed for the manipulative case.

Labels

List of frames and associated labels

Relation: *IS*

Frame: Food

- FOOD_Food_LU: Words referring to food items (i.e., LUs in the Food frame).
- FOOD_Constituent_Part: A component or a part of the FOOD_LU.
- FOOD_Descriptor: A characteristic or classification of the FOOD_LU, including both descriptive and typological information. It is further annotated for intrinsic polarity: (1) Positive [DESCR_Positive]; (2) Negative [DESCR_Negative]; (3) Neutral [DESCR_Neutral].
- FOOD_Count: The number of units in which the FOOD_LU is quantified.
- FOOD_Unit: The standard unit of measurement used to quantify the FOOD_LU (e.g., grams, liters).

Relation: *COMPOSED BY*

Frame: Ingredients

- INGR_Material_LU: The entity which is used to make a Product.
- INGR_Food_Product: The entity or substance that is created out of the Material, in this case the food.

Relation: *CREATED BY/ CREATED WITH*

Frame: Cooking_Creation

- COOKING_CREATION_LU Every word which expresses the action of cooking.
- COOKING_CREATION_Cook The person who prepares the Produced_Food.
- COOKING_CREATION_Produced_Food It is the result of a Cook's efforts.

Frame: Preserving

- PR_LU Every word expressing the act of preserving something from decaying.
- PR_Agent The person performing the intentional act that leads to the preservation.
- PR_Medium The substance in which the Patient might be submerged or treated with to be preserved.
- PR_Food_Patient The organic or non-organic matter that undergoes preservation (in this case, food).

Relation: *USED AS/USED FOR/MEANT TO*

Frame: Using_Cure

- CURE_LU: Every word expressing the act of treating or curing someone.
- CURE_Affliction: It represents the injury, disease, condition, or pain felt by the patient.
- CURE_Healer: It represents anyone who treats or cures the Patient.
- CURE_Food_Treatment The ingested, applied, injected, etc. substance designed to cure the Patient.
- CURE_Patient It is the sufferer of the injury, condition, disease or pain.

Frame: Using_Preserving

- PR_LU Every word expressing the act of preserving something from decaying.
- PR_Agent The person performing the intentional act that leads to the preservation.
- PR_Food_Medium The substance in which the Patient might be submerged or treated with to be preserved (in this case food).
- PR_Patient The organic or non-organic matter that undergoes preservation.

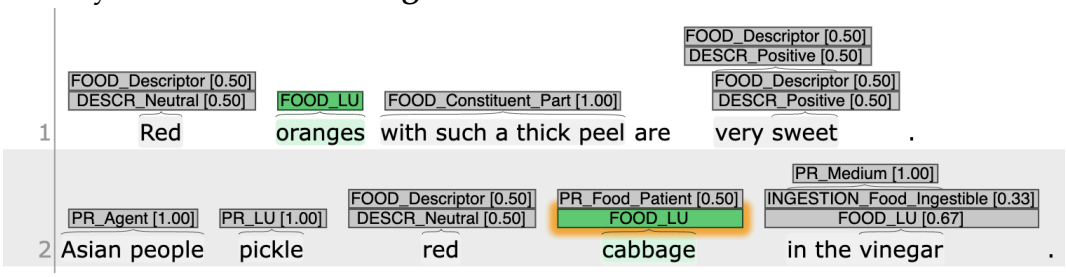
Frame: Using_Ingestion

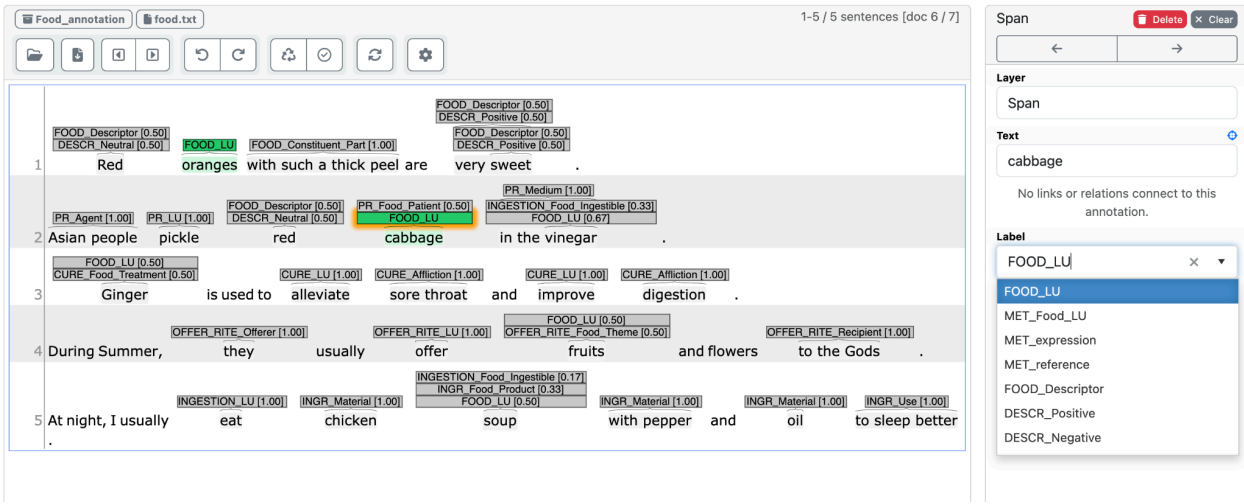
- INGESTION_LU Every word expressing the act of eating or drinking.

- INGESTION_Food_Ingestible The Ingestibles are the entities that are being consumed by the Ingestor.
- INGESTION_Ingestor The Ingestor is the person eating or drinking

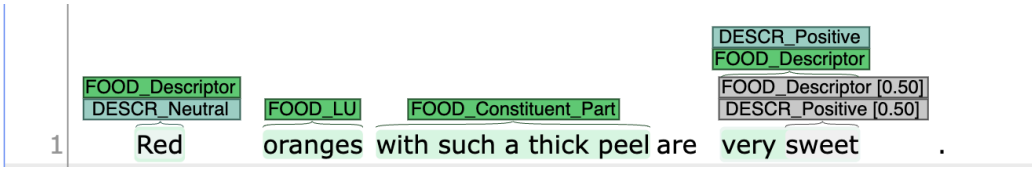
Annotation Workflow

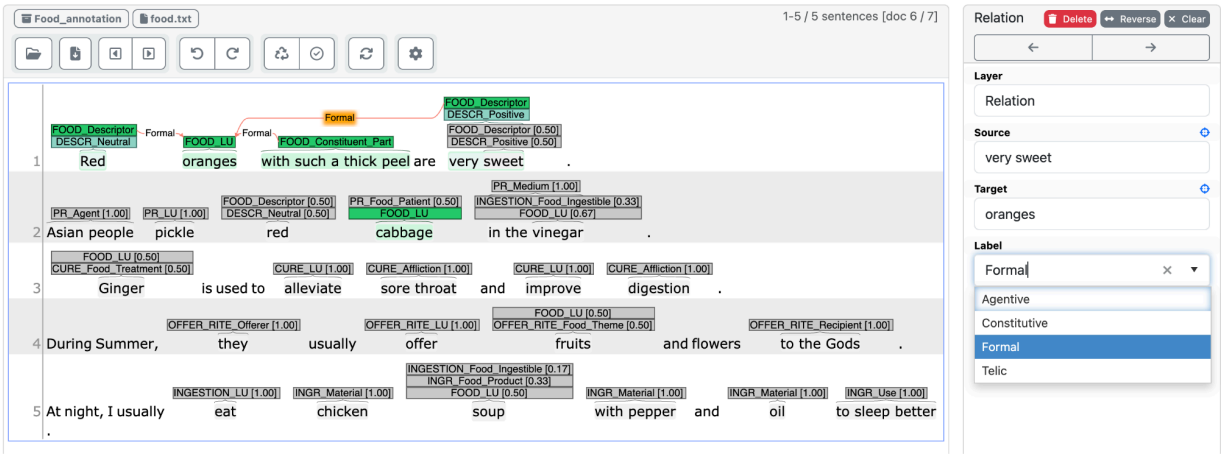
1) In a given sentence, the annotator has to:
   - Identify the **words referring to food** and annotate them as FOOD_LU.



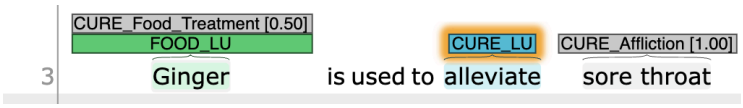


   - Annotate **Frame Elements (FEs)** related to each FOOD_LU in the **FOOD frame**, and link them to the LU using the relation *formal*.

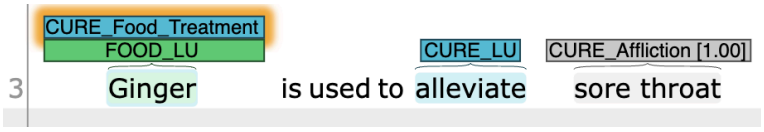




2) Check whether the food item **participates in another event or usage** that activates another frame (e.g., CURE, INGESTION, PRESERVING), namely if any other LUs or food realizations is present . If so:
   - Annotate the predicate (e.g., *alleviates*) or any other argument of other frames (e.g., *chicken* as Ingredient) as the LU of that frame, e.g., 'alleviates' as CURE_LU.



   - Re-annotate the food item with its new role in the frame (e.g., CURE_Food_Treatment for *ginger*).



More on FrameNet

Qualia Relations in FrameNet Brasil

Frame Semantics already embody a methodology whose aim is to represent the meaning of words especially through their context. Nevertheless, the representation of food in FrameNet is limited to considering food merely as an entity that can be ingested. But there are characterizations of food that do not entail its ingestion, as well as purposes that extend beyond mere consumption as a source of nutrition. For instance, when the purpose of food is clearly stated as a medicinal tool more than a nutritional item, which sometimes can entail not ingesting the considered food (e.g., a piece of meat applied on a bruise). To deal with this limitation, we resorted to the characterization of the different dimensions of a lexical item as presented in FrameNet Brasil. In this framework, the authors describe each lexical item through 4 main dimensions, named qualia roles. Each qualia role represents the different dimensions in which the meaning of a lexical item may be characterized.

The qualia structure encodes semantic dimensions essential to meaning, through four roles:

Formal (entity IS): describes the basic category for the item and provides the information that distinguishes an entity within a larger set inside its semantic domain (e.g., banana is a food);

Example: Ginger Ale **is** a plant-based drink.

Constitutive (entity IS COMPOSED BY): expresses a variety of relations concerning the internal constitution of an entity (e.g., banana is formed by banana peel, pulp, etc.);

Example: Ginger ale **is made of** carbonated water, ginger, and sugar.

Agentive/Manipulative (entity IS CREATED BY/WITH): concerns the origin of an entity, its creator or its coming into being, every possible agent or methodology that led to the creation of the food entity we are considering (e.g., banana cake created by a cook, created by baking bananas, flour etc.)

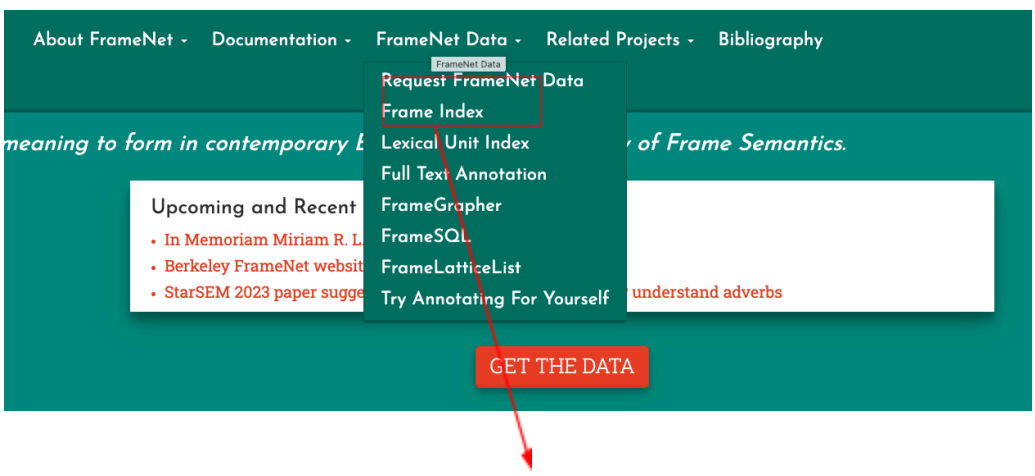
Example: The chef **prepared** the ginger ale using sparkling water.

Telic (entity IS MEANT TO/USED AS/USED FOR): concerns the typical function or purpose of an entity, i.e., what the entity is meant to or used for (e.g., banana used as medicine).

Example: Ginger ale is consumed **<u>to soothe</u>** nausea.

Consulting Frames in FrameNet

In FrameNet, there is a collection of nearly 1,200 frames for the English language. The repository is structured in such a way that it is possible to consult both the list of lexical units for the frames contained, and the frames themselves with their descriptions, list of LUs, and frame elements. As we have borrowed and re-adapted frames from FrameNet, we provide a brief overview of the type of information available in the repository. For those wishing to examine the frame element descriptions in greater detail and trace their sources, the repository can be consulted directly. If you are searching for a specific Lexical Unit and wish to identify the frames that contain it, you can refer to the Lexical Unit index. By entering the desired word in the search bar, you will be presented with the frames in which it occurs.



FrameNet Index of Lexical Units

This page is an index to alphabetical lists of the names of the lexical units (LUs).

Each LU name is followed by the part of speech, the name of the relevant frame, and its status. If a le
and the annotated sentences. Lexical units on which work has not been completed may have only a lin

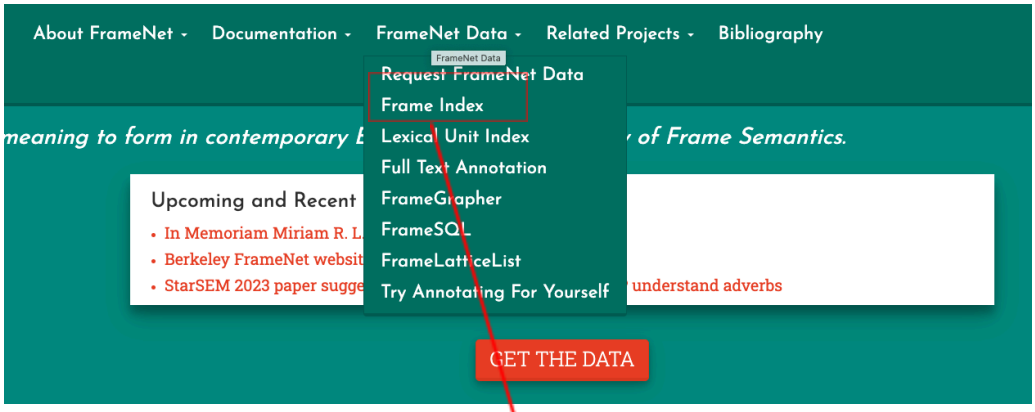
meat Search

| # | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | All |

Search: meat

- meat.n (Food) **Finished_Initial** Lexical Entry Annotation

Alternatively, if you wish to examine a specific frame along with its description and the frames it is associated with, you can simply search for it in the frame list.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of Frame Semantics.
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In Memoriam Miriam R. L
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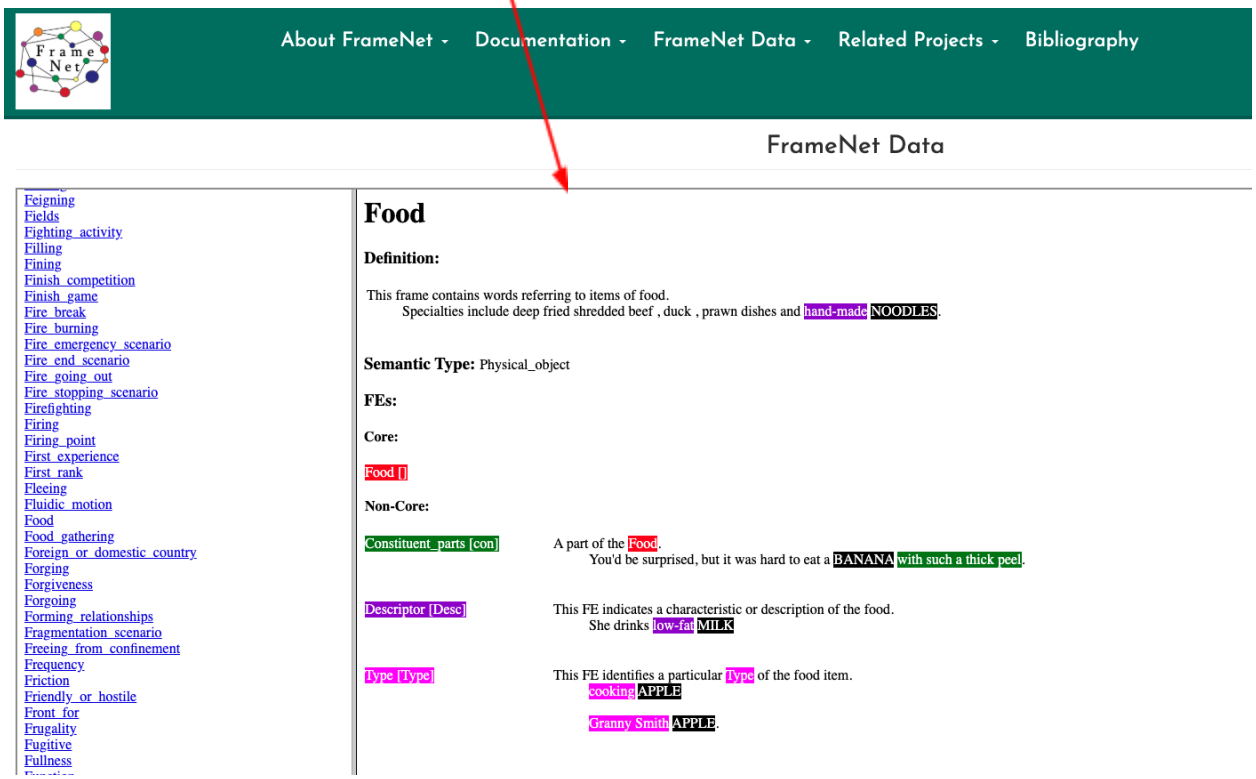
understand adverbs
GET THE DATA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Food
Definition:
This frame contains words referring to items of food.
Specialties include deep fried shredded beef , duck , prawn dishes and hand-made NOODLES.
Semantic Type: Physical_object
FEs:
Core:
Food []
Non-Core:
Constituent_parts [con]
A part of the Food.
You'd be surprised, but it was hard to eat a BANANA with such a thick peel.
Descriptor [Desc]
This FE indicates a characteristic or description of the food.
She drinks low-fat MILK
Type [Type]
This FE identifies a particular Type of the food item.
cooking APPLE
Granny Smith APPLE.